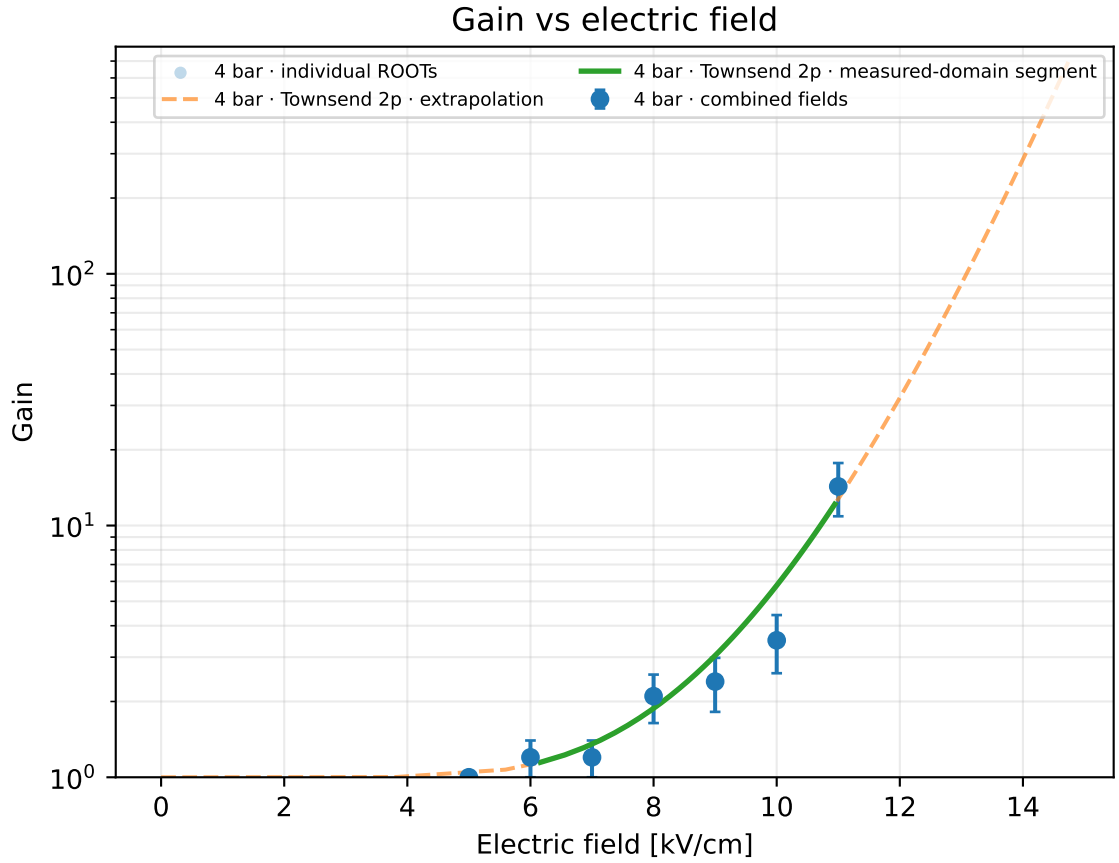
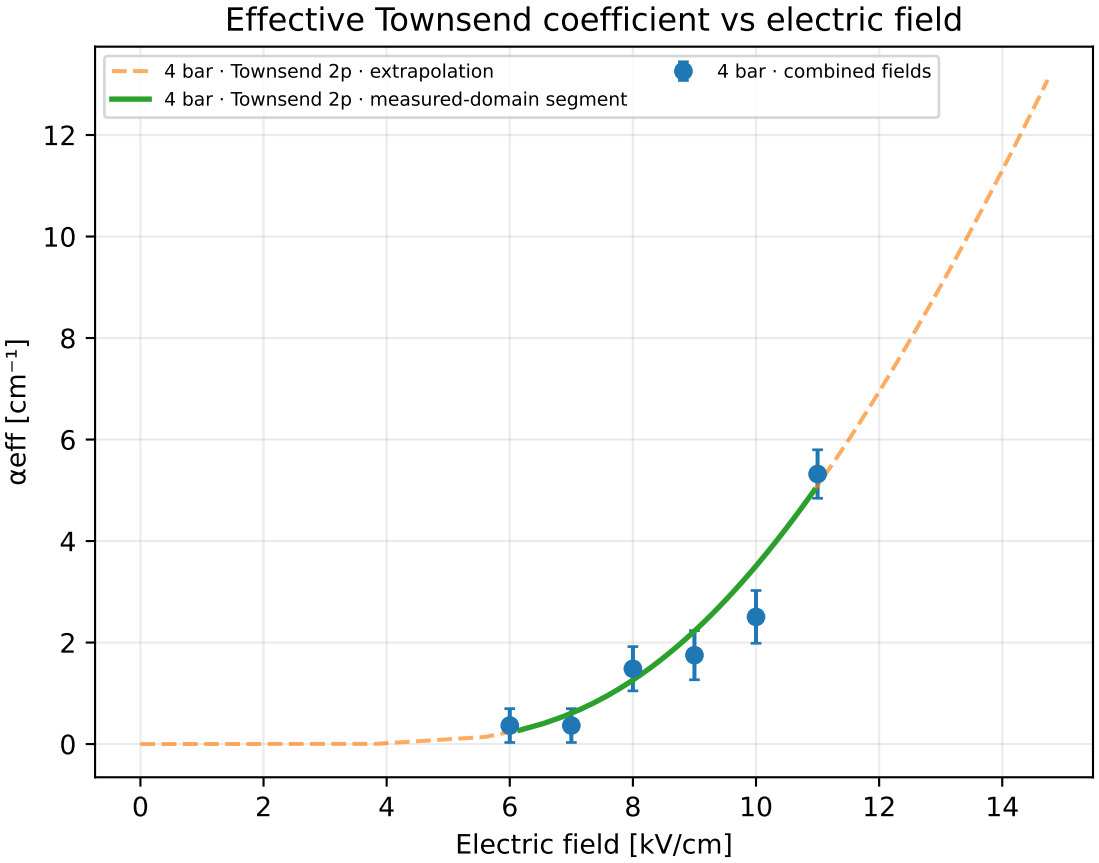
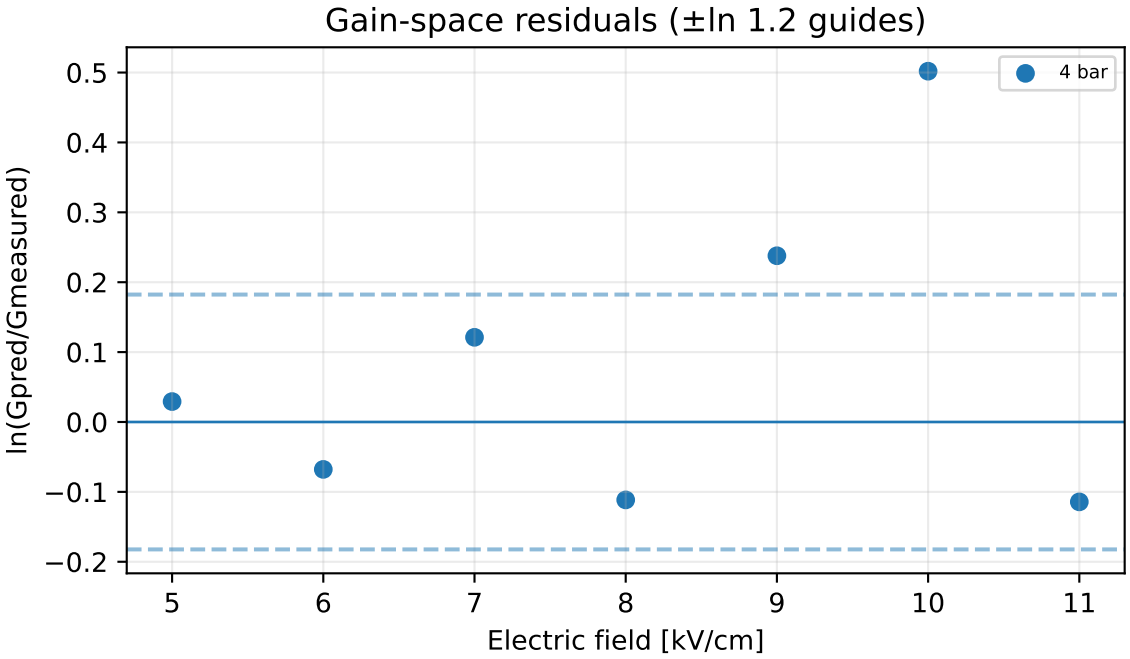
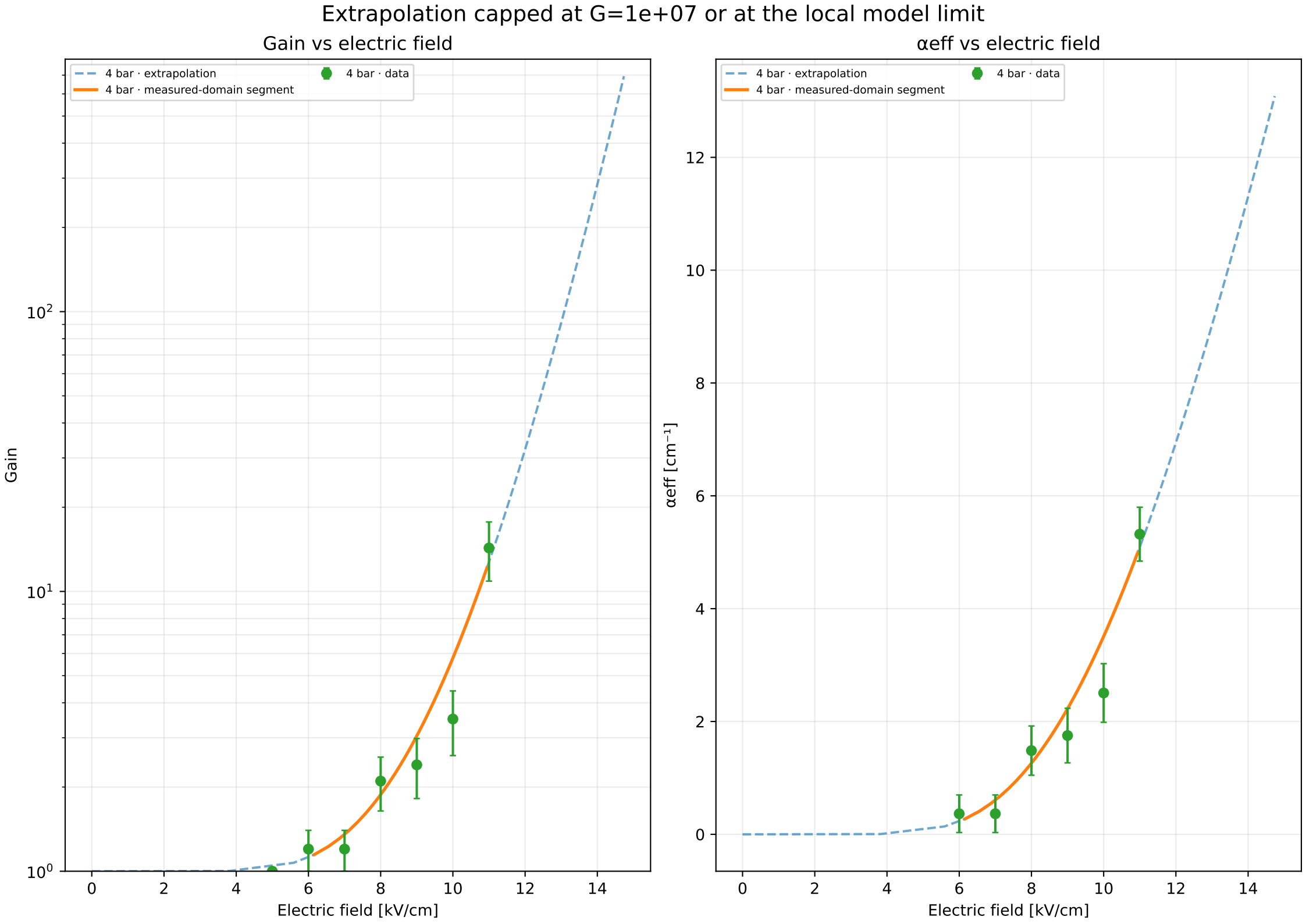
ArXe · Ar 99 % / Xe 1 % · gap 5 mm








```
ROOT runs: 7
Unique physical points: 7
Pressures: 4 bar
Selected model: Townsend 2p
Status: rejected_large_training_residuals
Training RMSE: ×1.399
Cross-validation median: ×1.274
Cross-validation worst: ×2.679
Reduced-field range: 1.5–2.75 kV cm⁻¹ bar⁻¹
A=52.645, B=10.236, m=0, n=1
Covariance condition: 885
Warnings: large_training_residuals
4 bar display limit: G=694 (local inversion limit)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
4 bar · extrapolation
4 bar · measured-domain segment
4 bar · data
Gain
Electric field [kV/cm]
αeff vs electric field
4 bar · extrapolation
4 bar · measured-domain segment
4 bar · data
αeff [cm⁻¹]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
4	5	1.25	1	0	—	10	1
4	6	1.5	1.2	0.2	0.364643	10	1
4	7	1.75	1.2	0.2	0.364643	10	1
4	8	2	2.1	0.458	1.48387	10	1
4	9	2.25	2.4	0.581	1.75094	10	1
4	10	2.5	3.5	0.91	2.50553	10	1
4	11	2.75	14.3	3.42	5.32052	10	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-5.1	×1.27	×2.68	no	large training residuals